

ANIMOVE



```
P1.x=diag(c(0, 0.001, 0.001))
P1.y=diag(c(0, 0.001, 0.001))

splayPar(mov.model=~1, err.model=list(x=~errX, y=~errY), drift.no=1,
data=nfsNew, fixPar=c(NA, 1, NA, 1, NA, NA, NA, NA))

t <- crwMLE(mov.model=~1, err.model=list(x=~errX, y=~errY), drift.no=1,
data=nfsNew, coord=c("longitude", "latitude"), polar.coord=0,
Time.name="Time", initial.state=initial.drift,
fixPar=c(NA, 1, NA, 1, NA, NA, NA, NA),
control=list(maxit=2000, trace=1, REPORT=10),
```




December 2025

Trajectory Centered Analysis

*Net Squared Displacement, First-Passage Time,
Recursions*

Trajectory centered analysis

Information extracted from the geometry of tracks, without context.

To keep in mind before we start:

- Some methods are **scale sensitive and sensitive to sampling schedule**.
- For some methods data should be in an **equidistant projection** (distances are preserved).

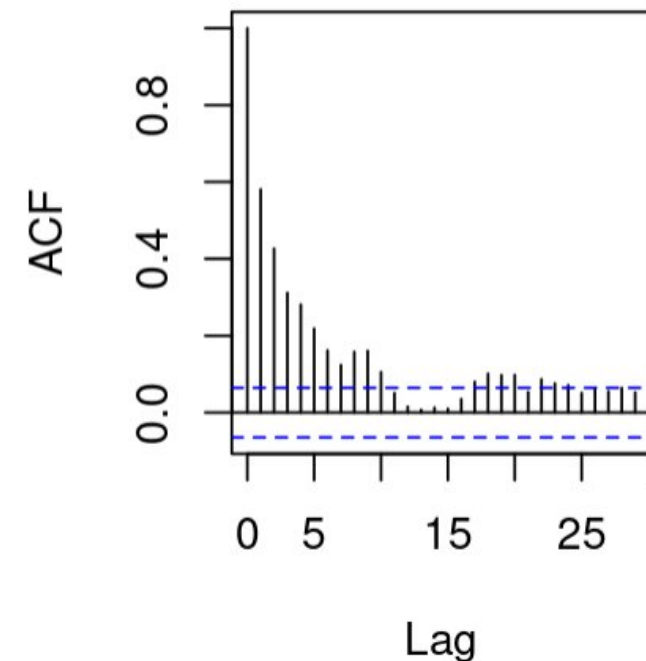
Auto-correlation

Auto-correlation is the **degree of correlation of a variable with itself over time**.

In almost all natural movements we can find some **degree of persistence in maintaining the direction and often also speed (correlation in geometric properties)**.

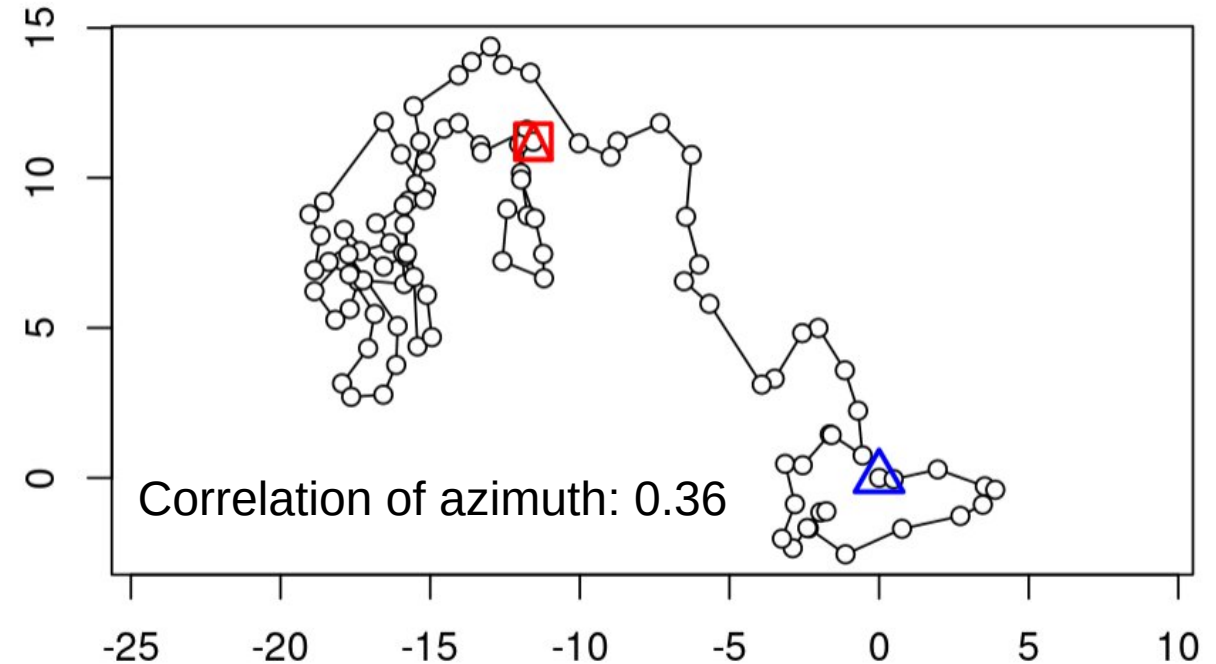
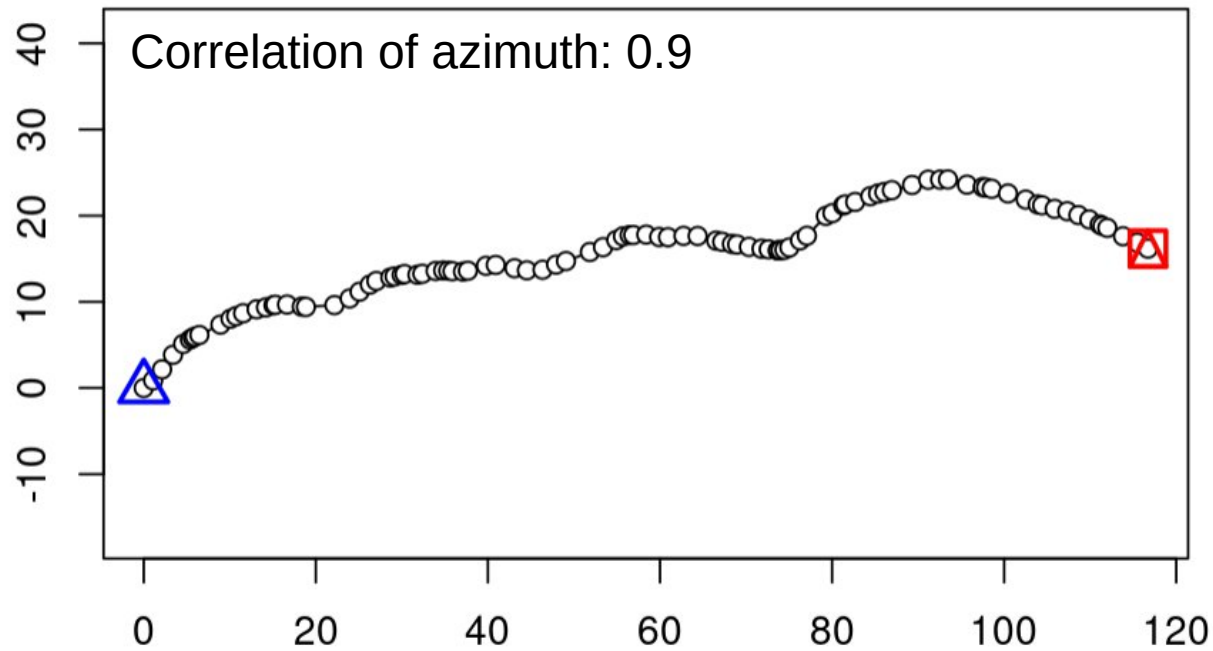
Auto-correlation is calculated and quantified based on correlation of values of a parameter (e.g. speed), dependent on the distance in the sequence of appearance (lag), between any two observations. Easy to explore in R using the function `acf(var)`.

Auto-correlation of speed in a fisher



Auto-correlation and movement process

The correlation structure can **inform about the underlying movement processes and changes in these processes** (directional vs foraging movement, periodicity, etc..).



Quantify movement process based on track geometry

Net square displacement (NSD):

- Quantifies the net squared distance traveled over time compared to a point of reference (calculates distance from each location to a point of reference)

First passage time (FPT):

- Assumes *regular sampling and equidistant projection*.
- Calculates the time it takes to cross a circle of a given radius, i.e. time the animal spends in a given area at a certain spatial scale ("draws" circles around each point and calculates how long it takes to leave the circle)

Recursions:

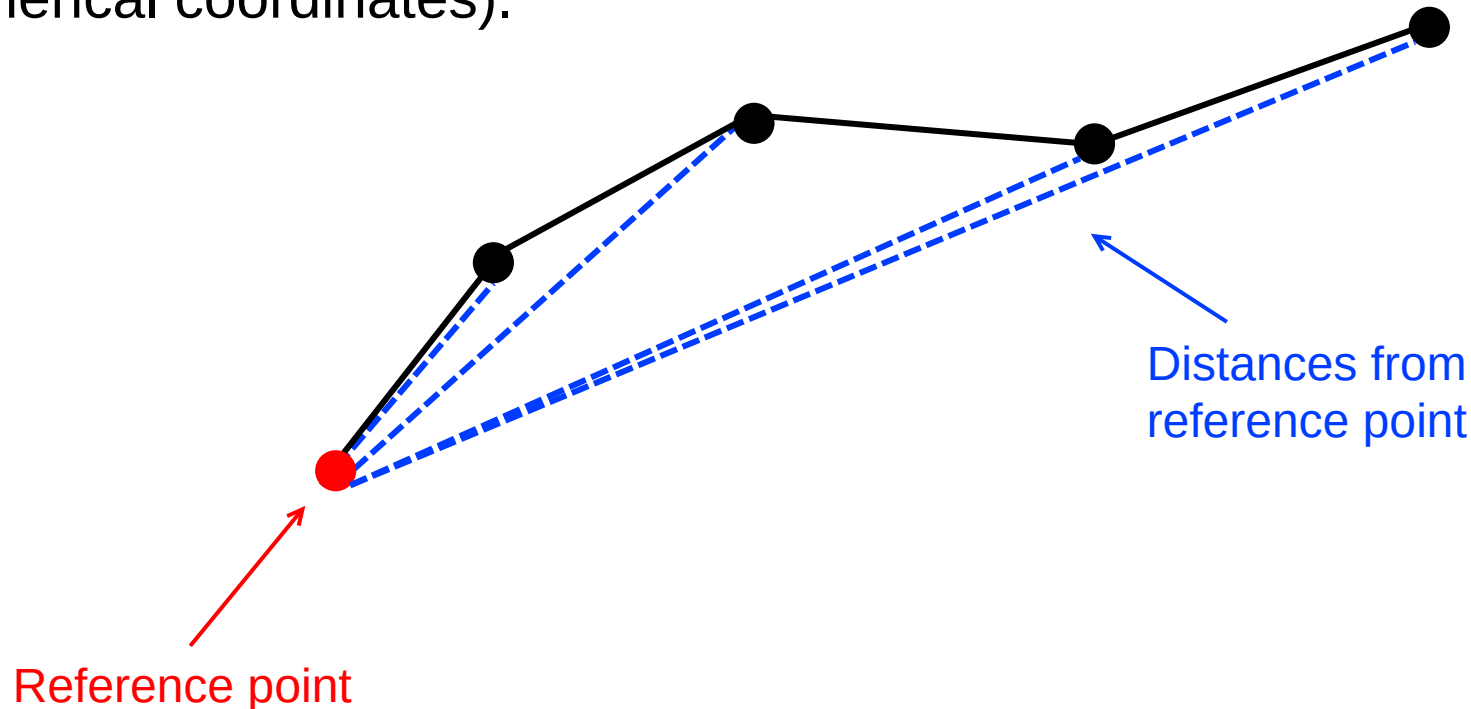
- Assumes *regular sampling and equidistant projection*.
- Identifies frequently-used locations across one or multiple individuals (by pre-specify locations of interest or by examining all trajectory locations)

Net-squared displacement (NSD)

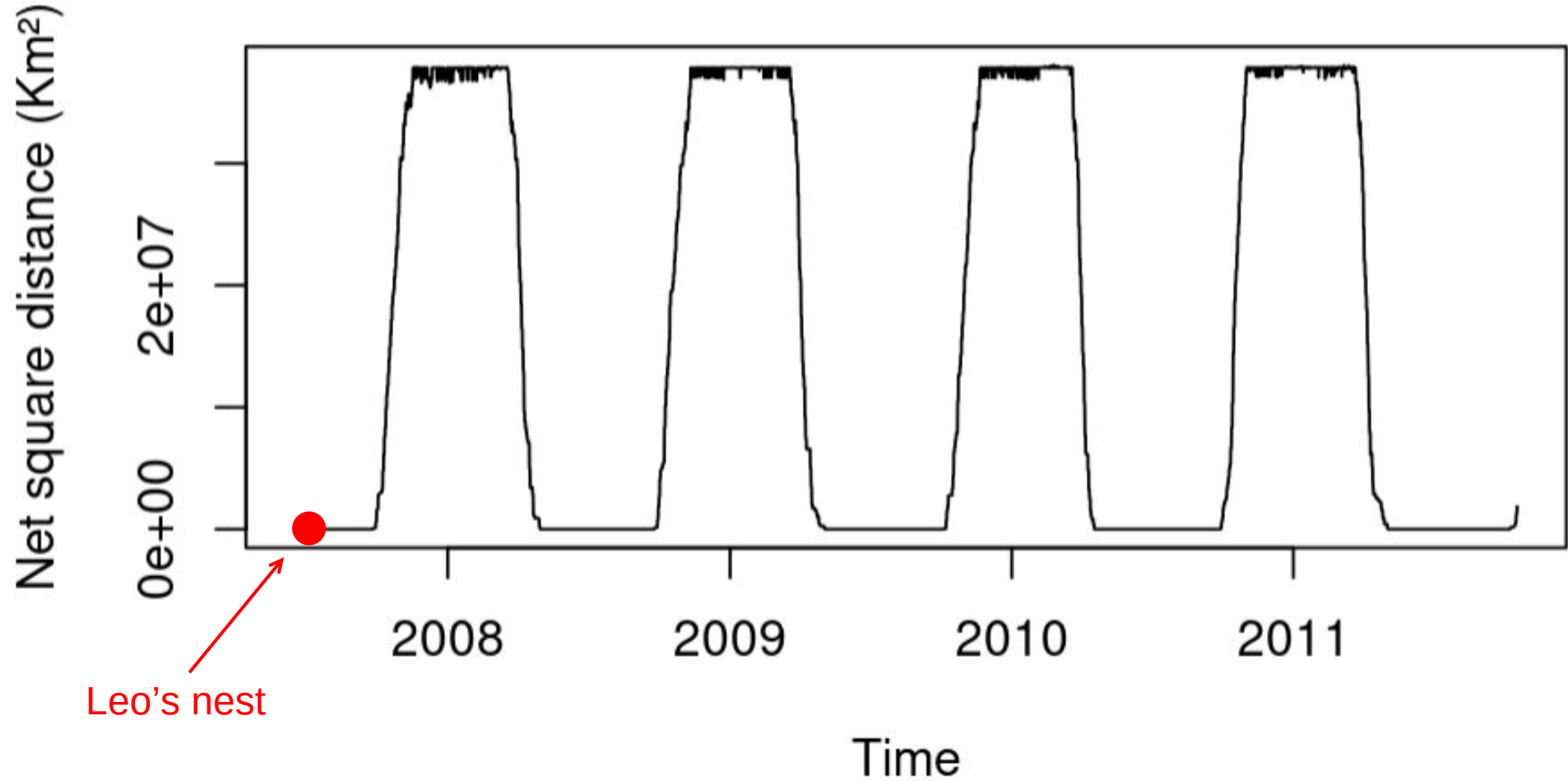
NSD quantifies the net squared distance traveled over time compared to a point of reference (calculates **distance from each location to the point of reference**).

A point of reference (nest, den, colony) is needed. If there is no point of reference, or animal is nomadic, use e.g. FPT (see later).

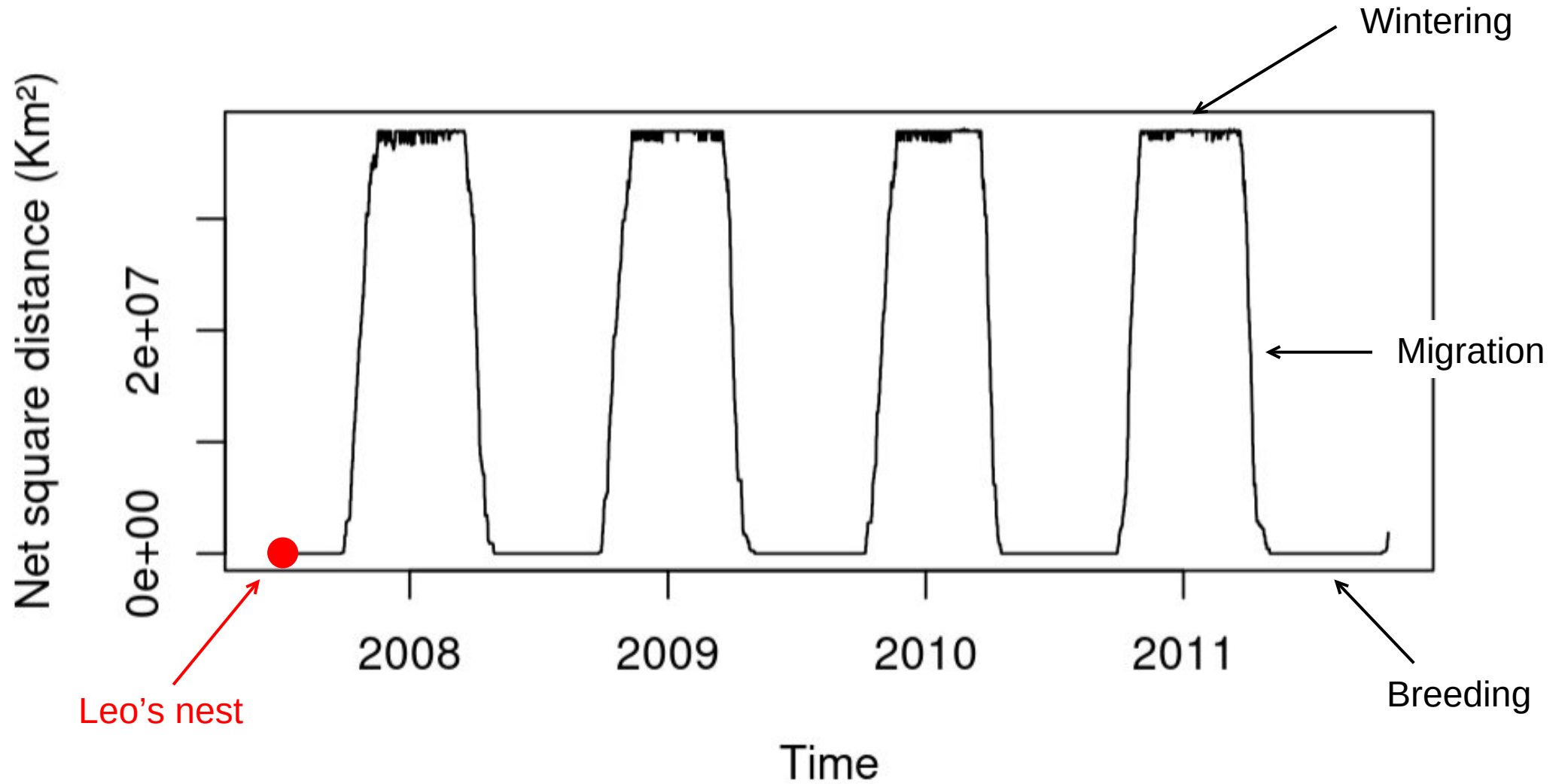
Implementation in any package calculating distances (e.g. `sf::st_distance`, which can also deal with spherical coordinates).



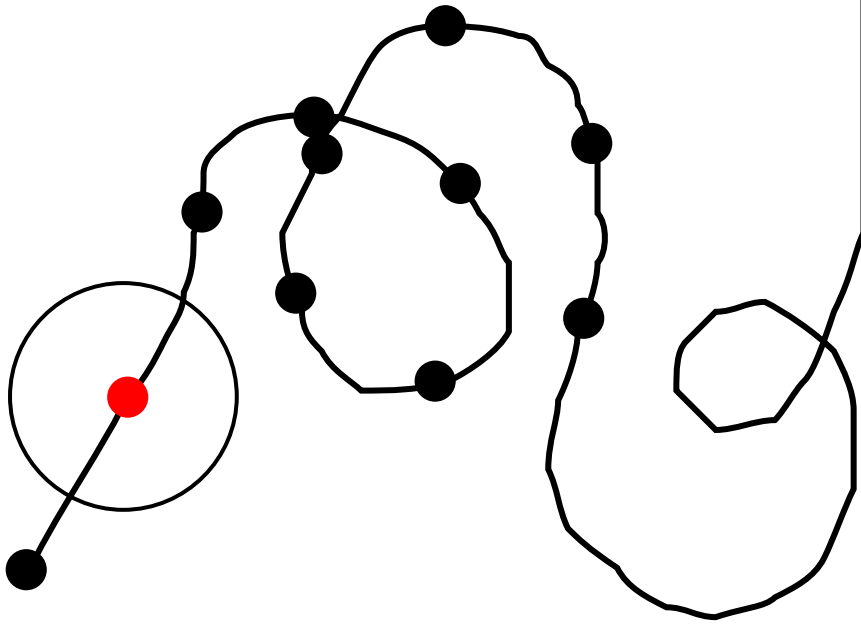
Net-squared displacement (NSD)



Net-squared displacement (NSD)



First-Passage Time (FPT)

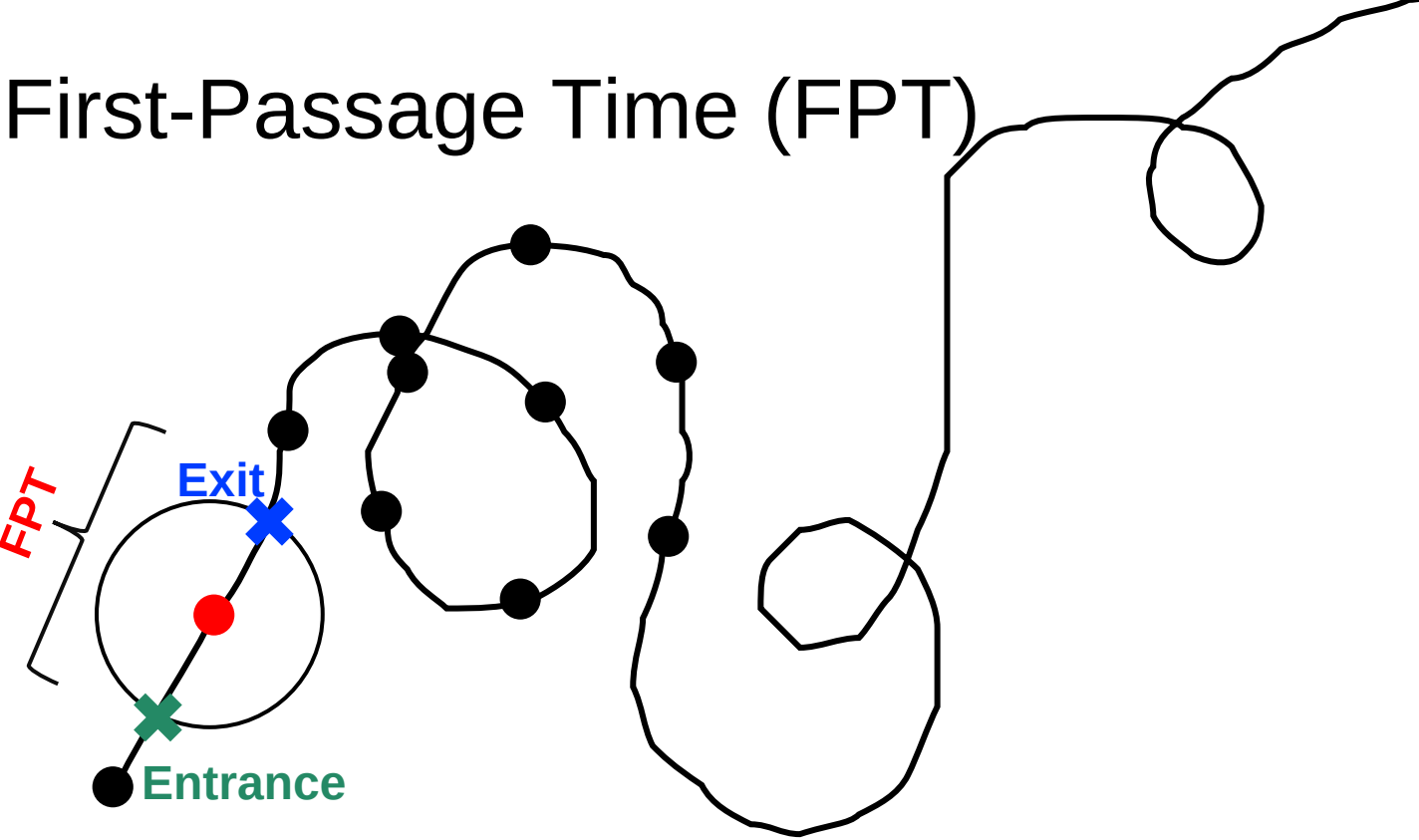


FPT calculates the **time it takes to cross a circle of a given radius**, i.e. time the animal spends in a given area at a certain spatial scale ("draws" circle around each point and calculates how long it takes to leave the circle).

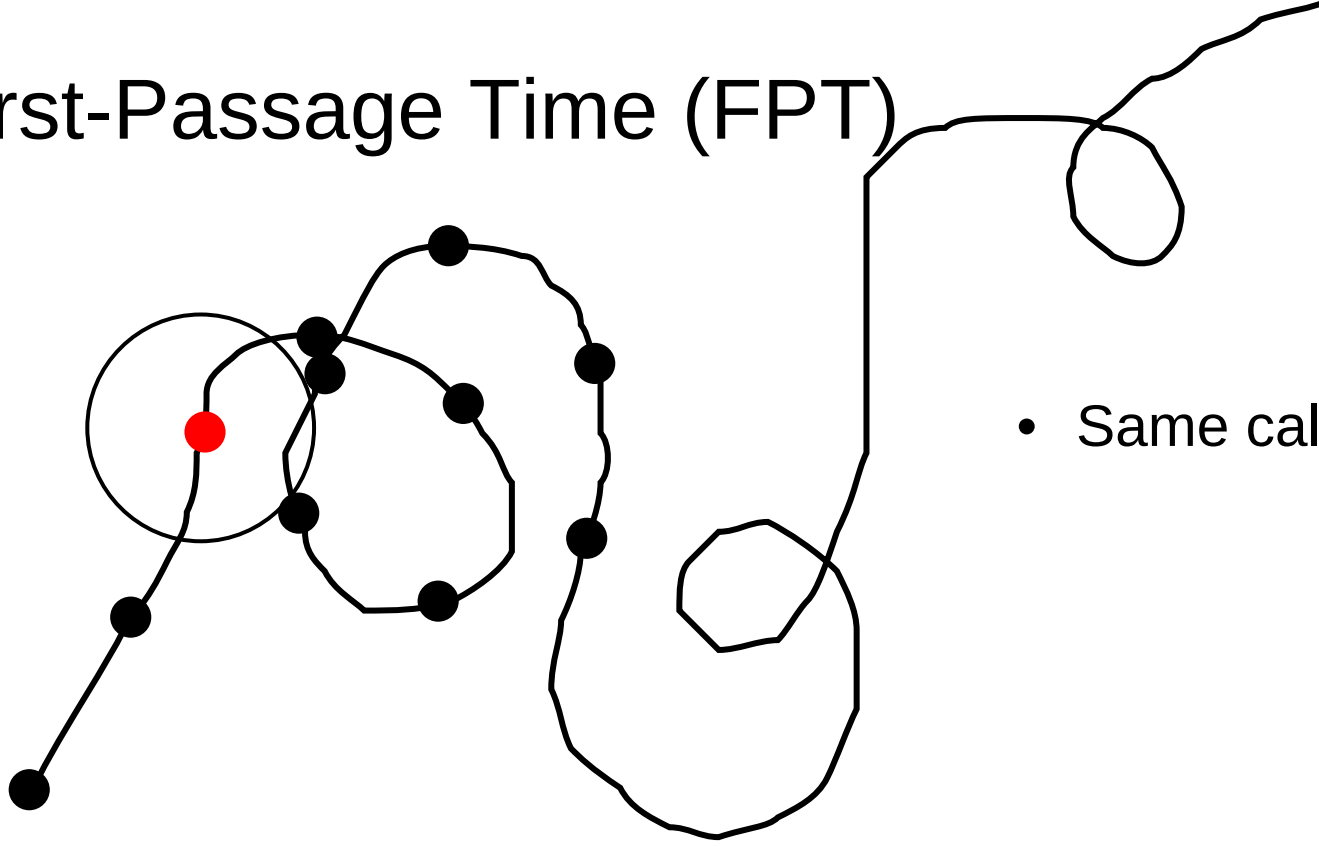
Implementation in R package `adehabitatLT`.

Requires regular sampling and equidistant projection.

First-Passage Time (FPT)

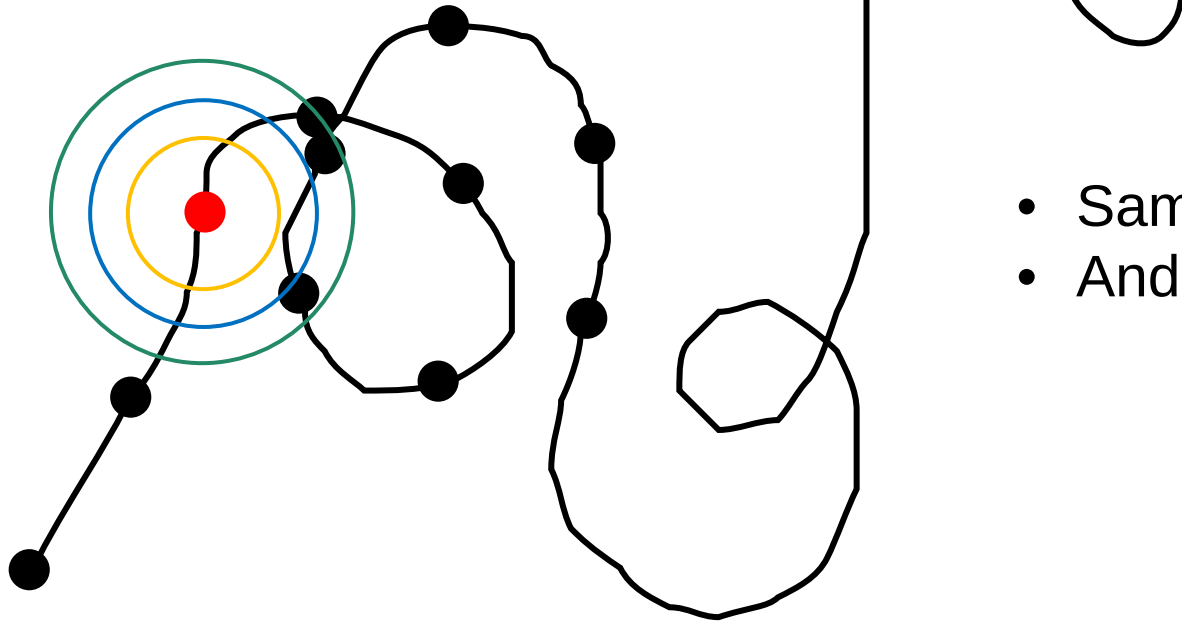


First-Passage Time (FPT)



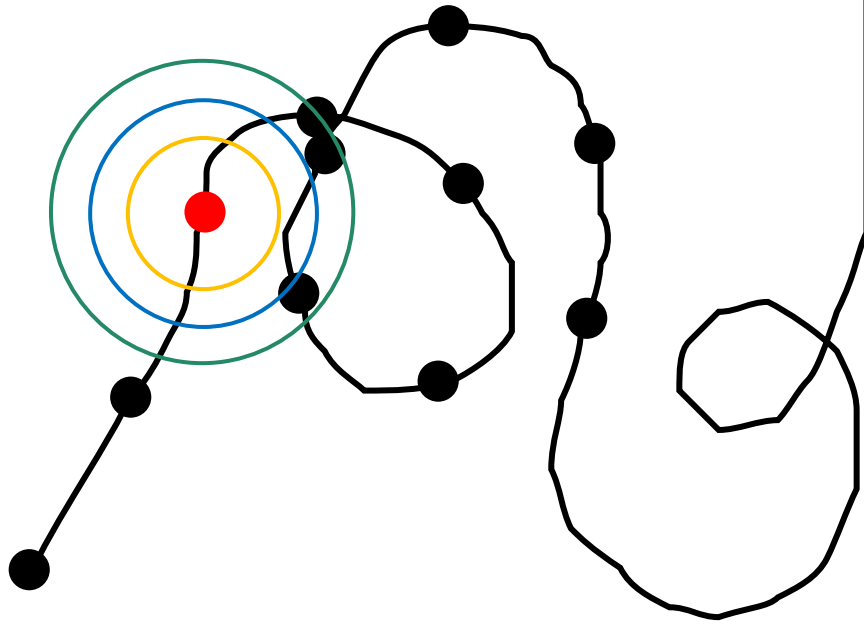
- Same calculation for each point in the trajectory

First-Passage Time (FPT)



- Same calculation for each point in the trajectory
- And for a sequence of different radii

First-Passage Time (FPT)

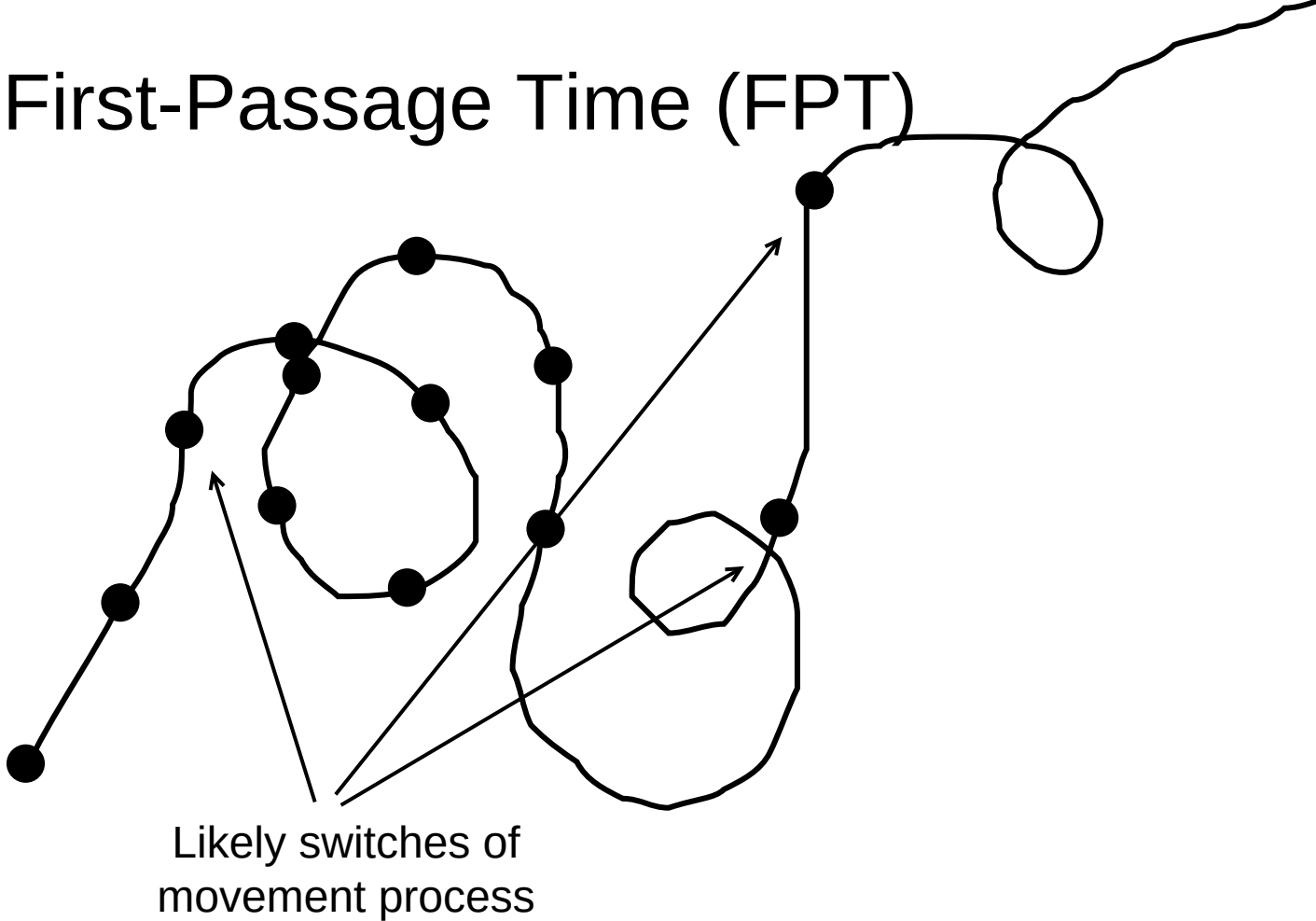


- Same calculation for each point in the trajectory
- And for a sequence of different radii

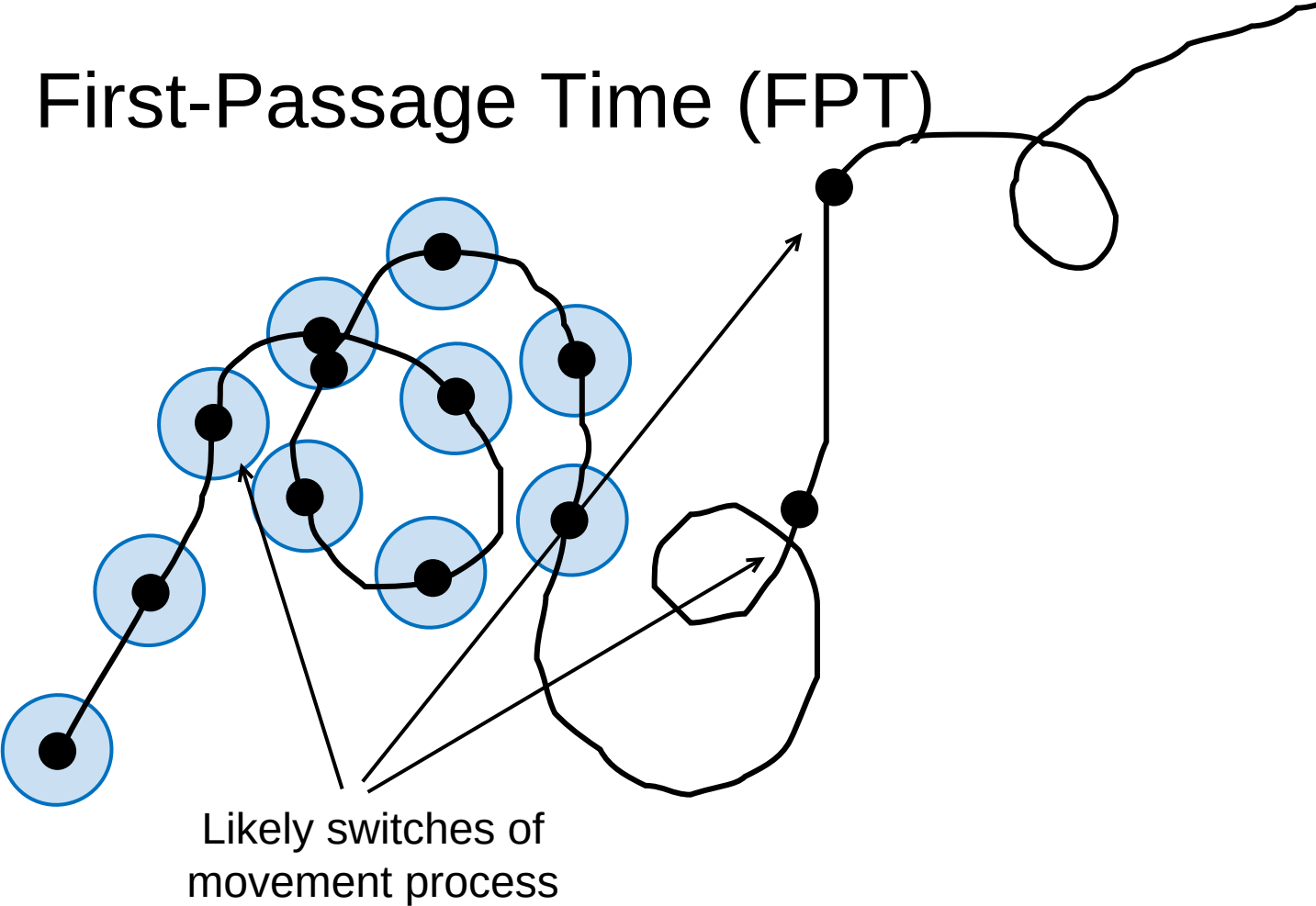
Will give us an understanding of the **movement processes that happen at different scales:**

- the **variance of $\log(\text{fpt}) \sim \log(\text{radii})$** can inform about the scales at which processes are likely to be changing
- the **slopes of the $\log(\text{meanFPT}) \sim \log(\text{radii})$** can indicate the type of movement at each scale. Flatter slopes indicate more directional movement, steeper slopes more brownian movement.

First-Passage Time (FPT)

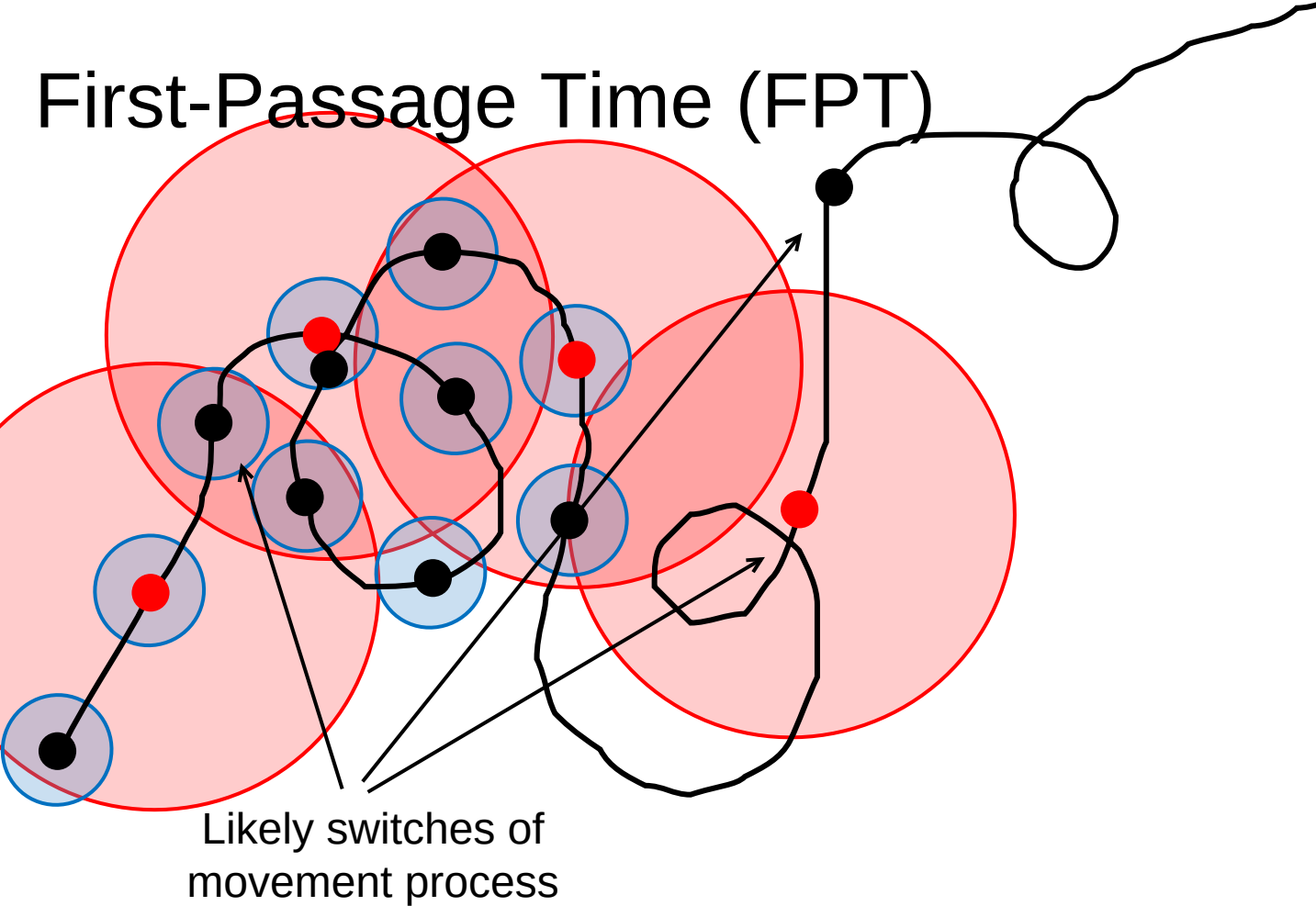


First-Passage Time (FPT)



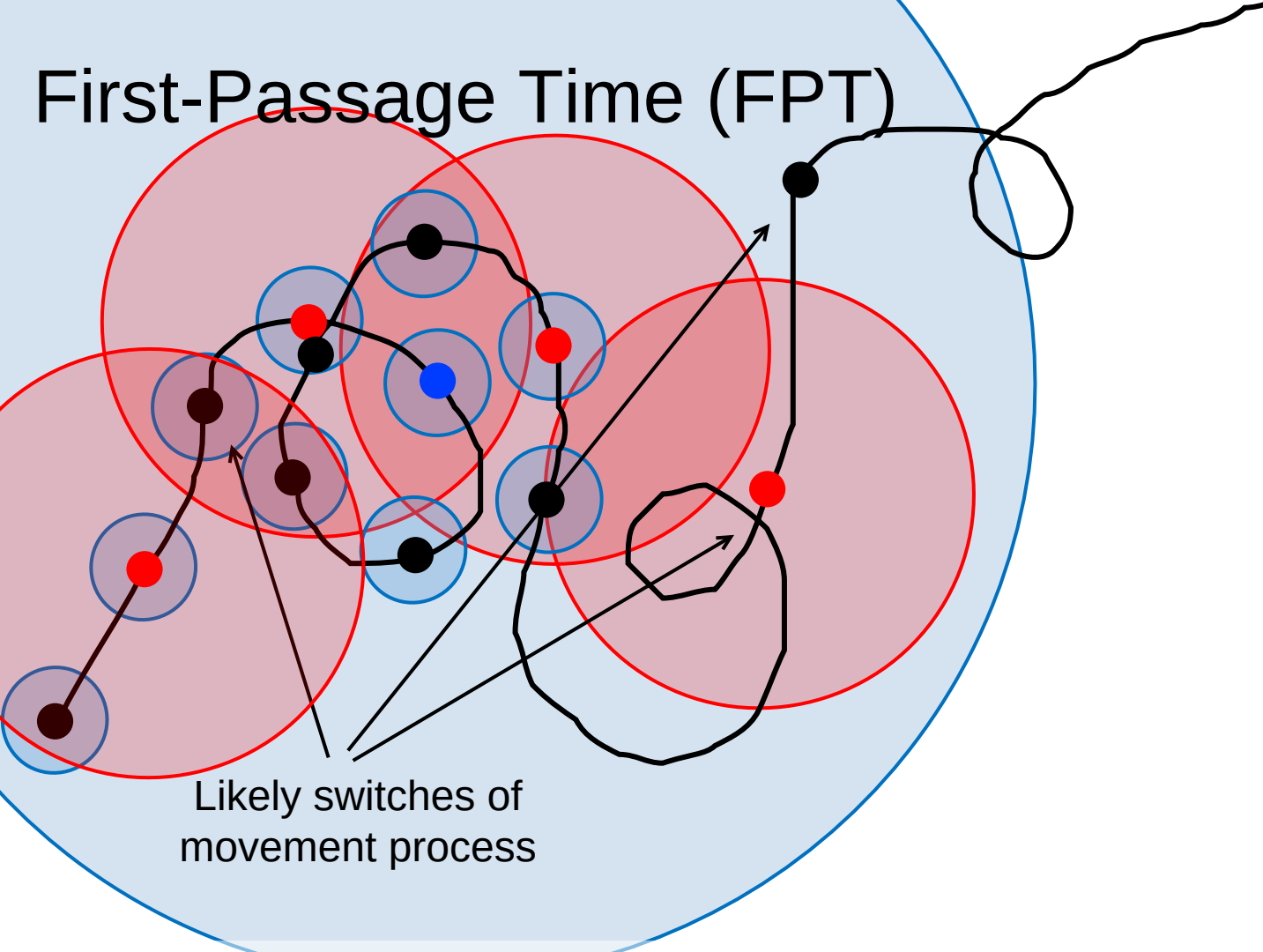
● Radii of low variance
(consistent movement process)

First-Passage Time (FPT)



-  Radii of high variance at the switch between behaviours
-  Radii of low variance (consistent movement process)

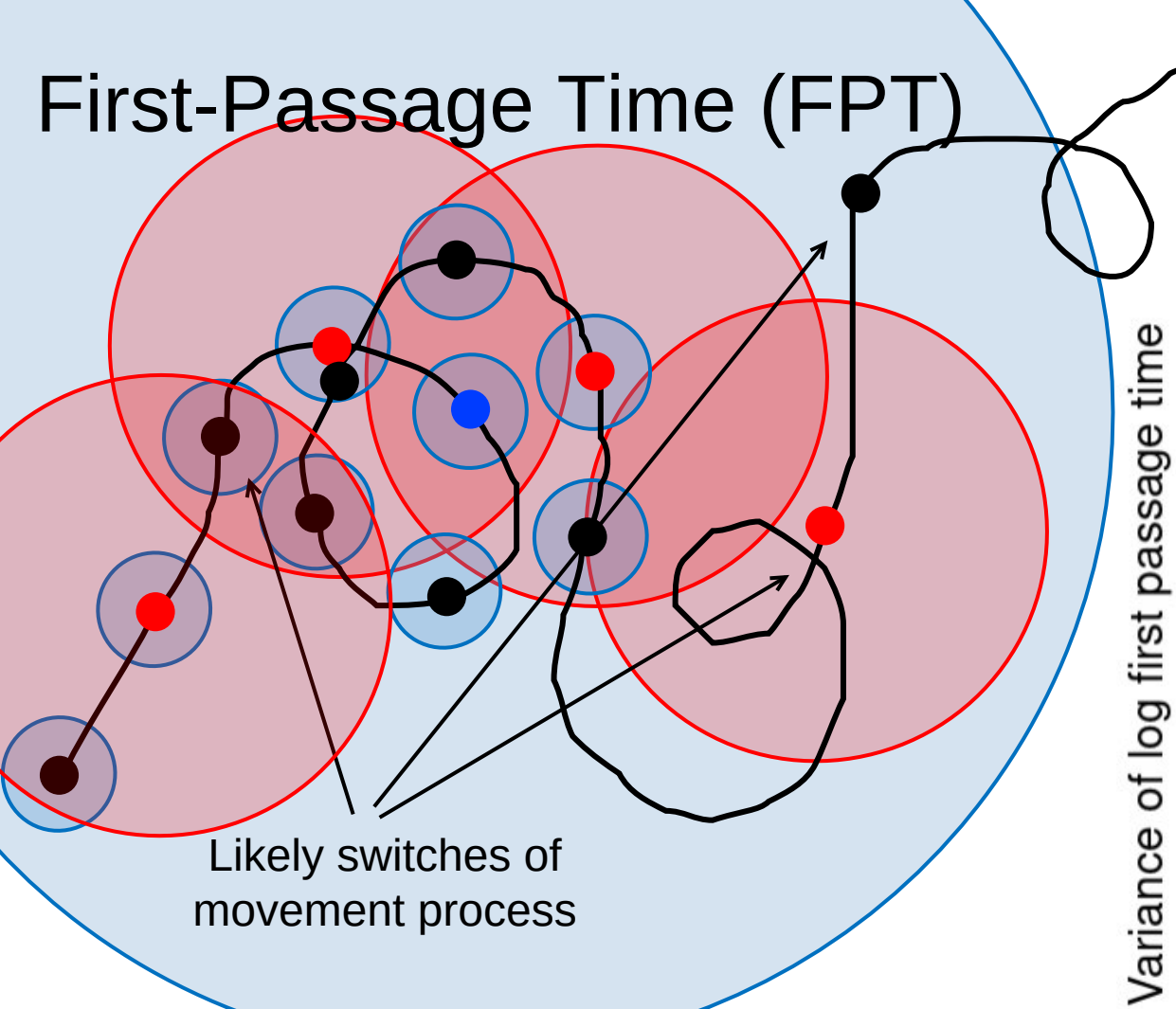
First-Passage Time (FPT)



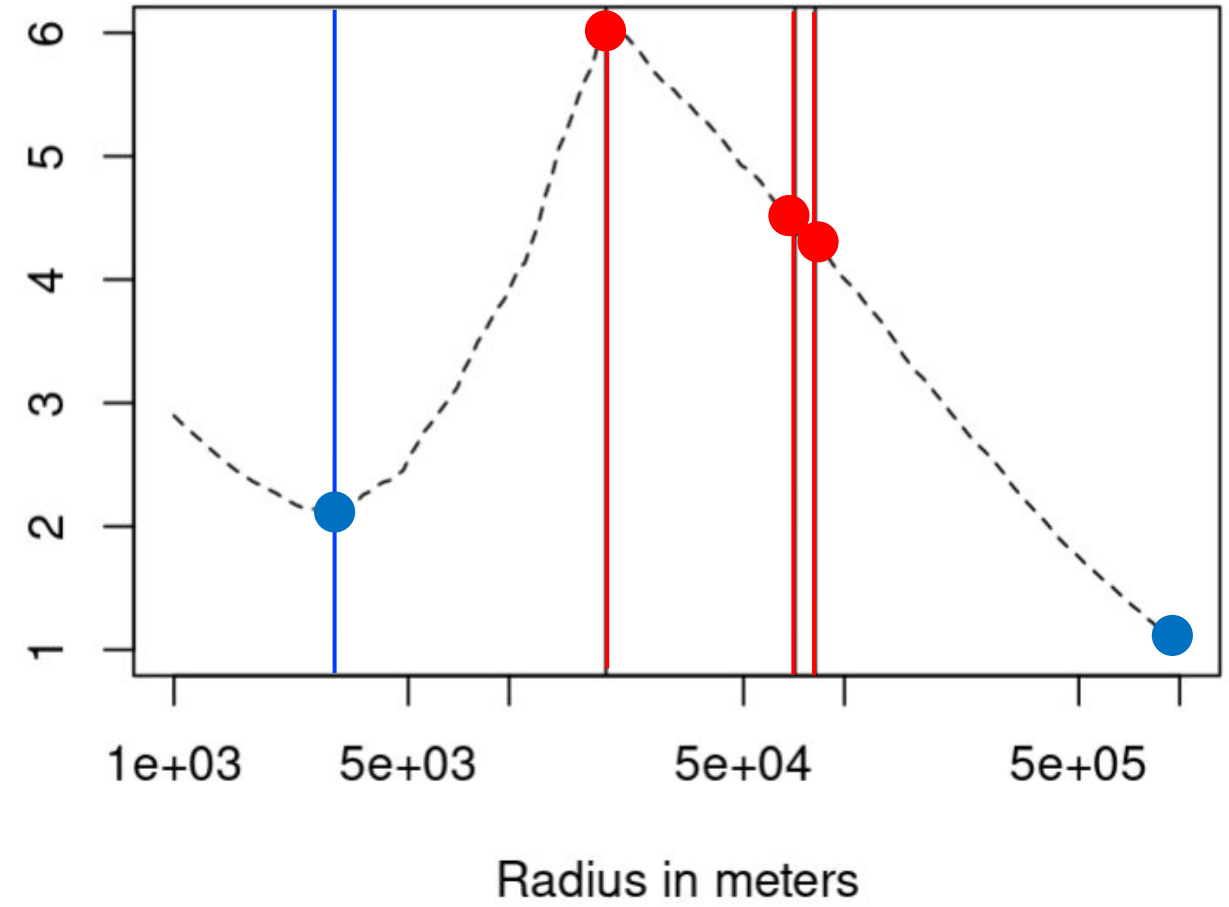
Likely switches of movement process

- Radii of high variance at the switch between behaviours
- Radii of low variance (consistent movement process)

First-Passage Time (FPT)



Variance of $\log(\text{FPT}) \sim \log(\text{radii})$

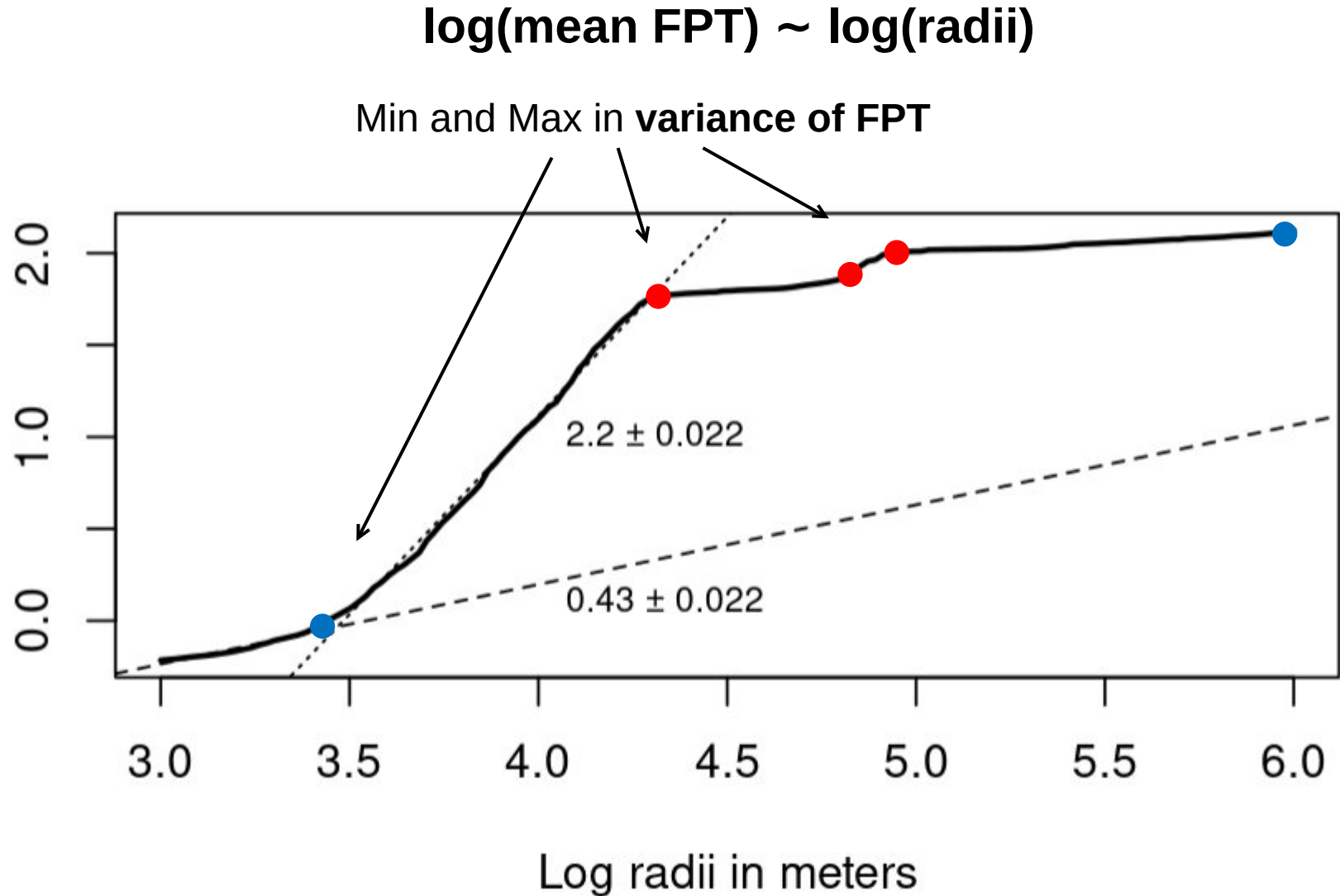


- Radii of high variance at the switch between behaviours
- Radii of low variance (consistent movement process)

First-Passage Time (FPT)

- **flat slope** (< 2):
directional movement
(larger radius = small
increase in time to cross it)
- **steep slope** (around 2):
brownian movement
(increase in radius = linear
increase in time to cross it)

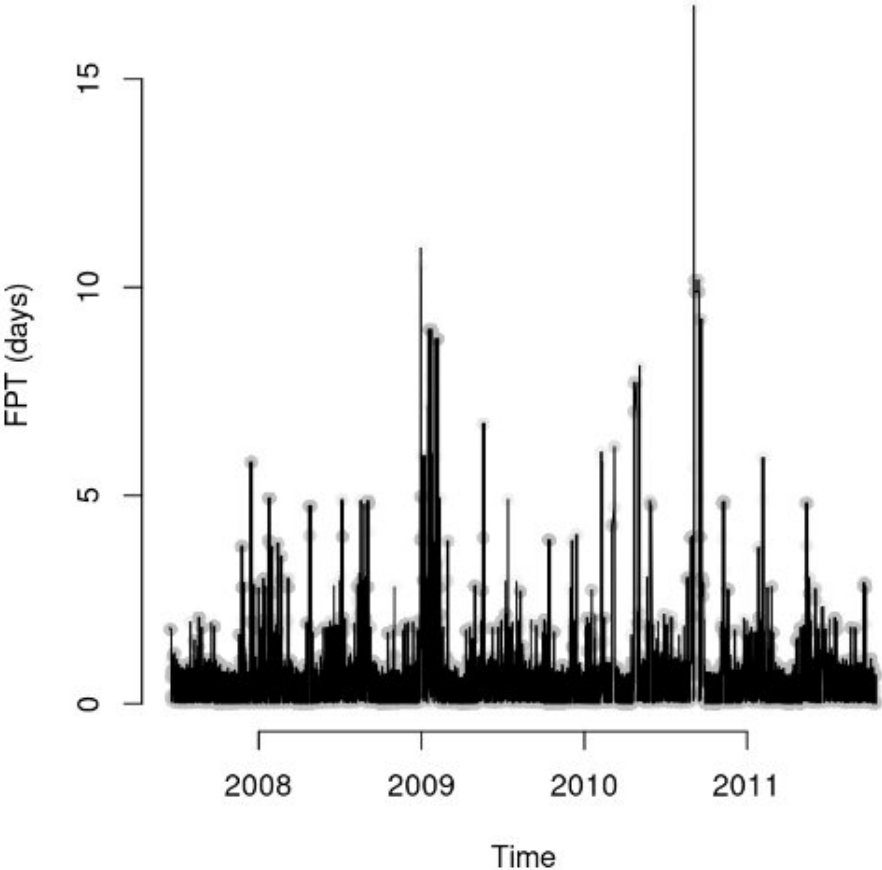
Log mean first passage time in days



First-Passage Time (FPT)

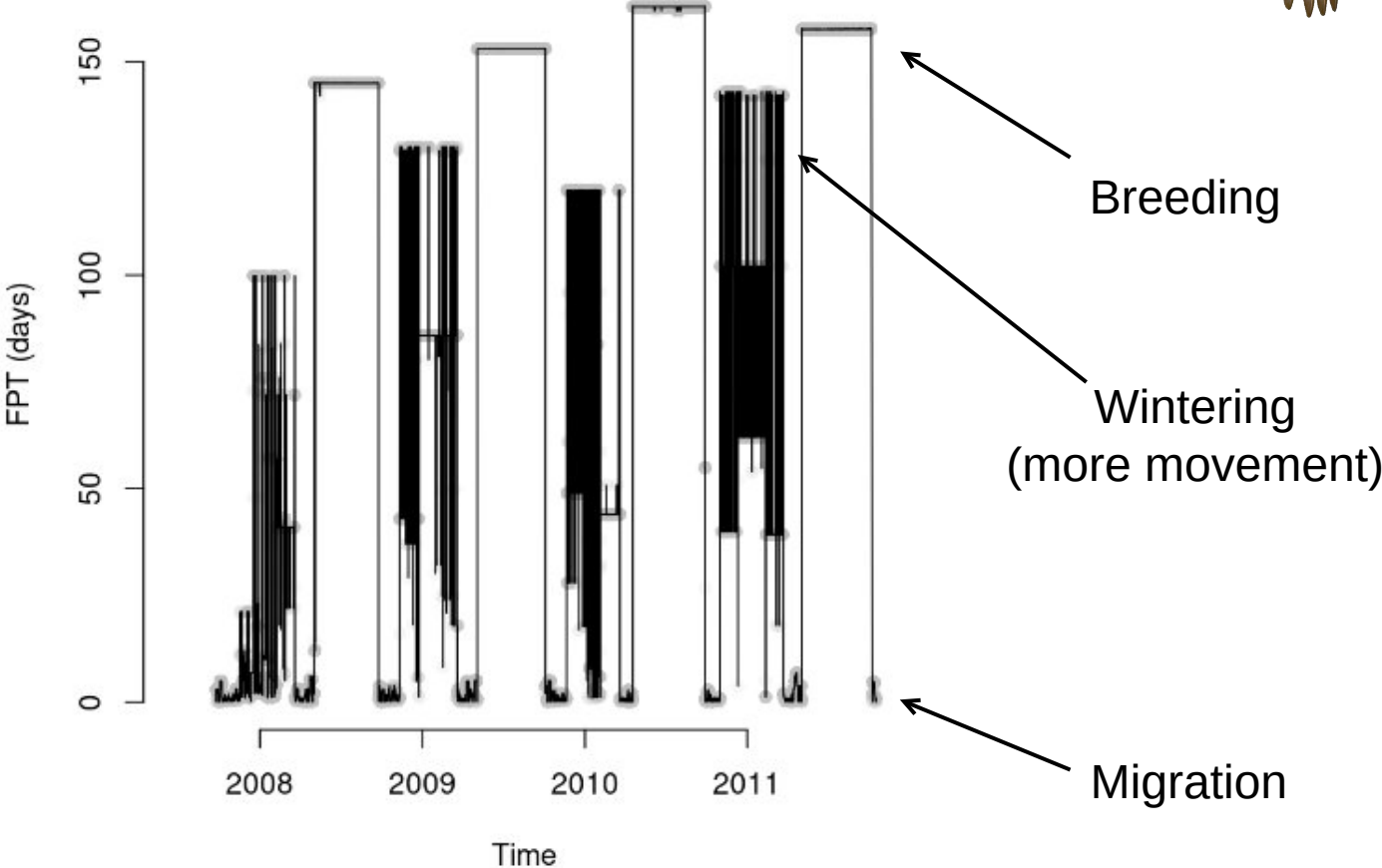


Radius 2773 meters



Radius ~3 km = probably daily ranges
(below 3 km Leo moves very directly)

Radius 71179 meters



Radius ~80 km = migration

Hands on

How to do this in R in the script:
`"6A_TrajectoryAnalysis_NSD_FPT.R"`

Recursions

Why do animals return?



Recursions

Applications of recursion analysis:

- Identify frequently-used locations across one or multiple individuals (by pre-specify locations of interest or by examining all trajectory locations)
- Visit-level analysis of **time of day, duration, time since last visit**, etc. and combine with other information (e.g, behavioral segmentation)
- **Intervisit** interval consistency at and across locations
- Visits to **user-specified polygon** (protected area, foraging ground)
- Residence time during user-specified intervals (**seasonal**, etc.)

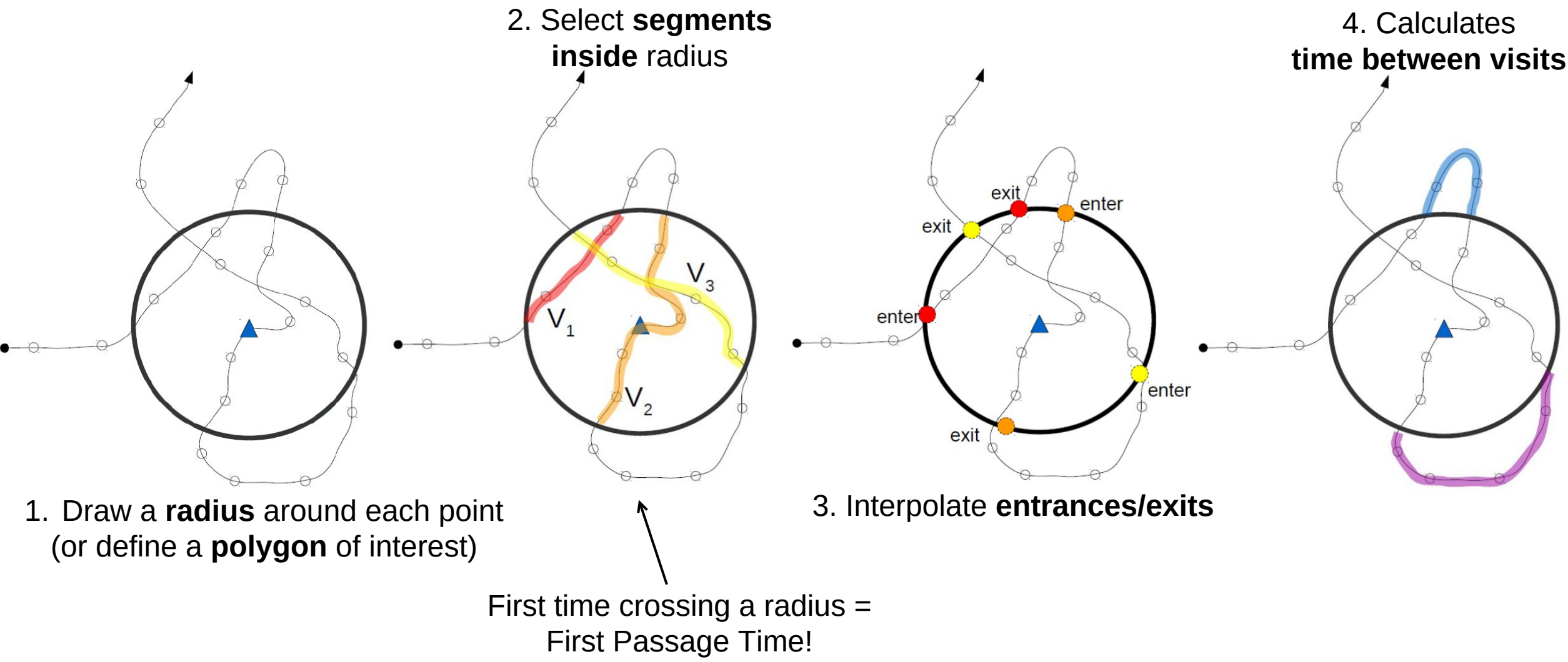
Analysis including environmental context:

- Spatially/temporally coincident visit **covariates** (e.g., NDVI, temperature, snow etc.)

Requires regular sampling.

Implementation in R package `recurse` (CRAN vignette by Chloe Bracis <https://cran.r-project.org/web/packages/recurse/vignettes/recurse.html>).

Recursions



Recursions



Number of revisits on a 500 m radius
around each location



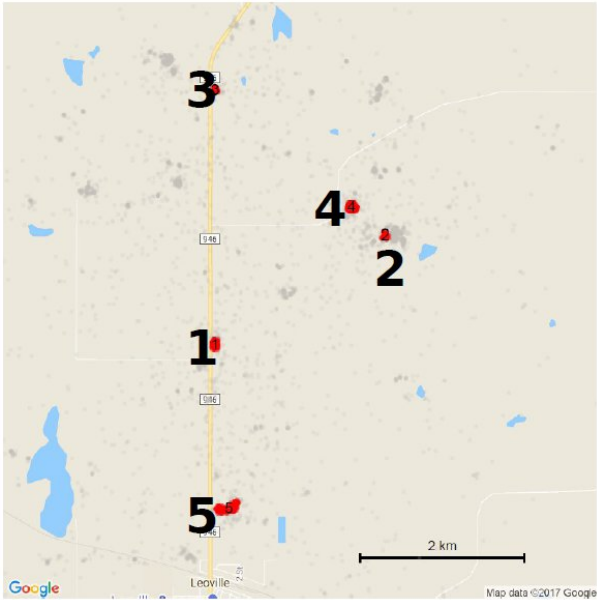
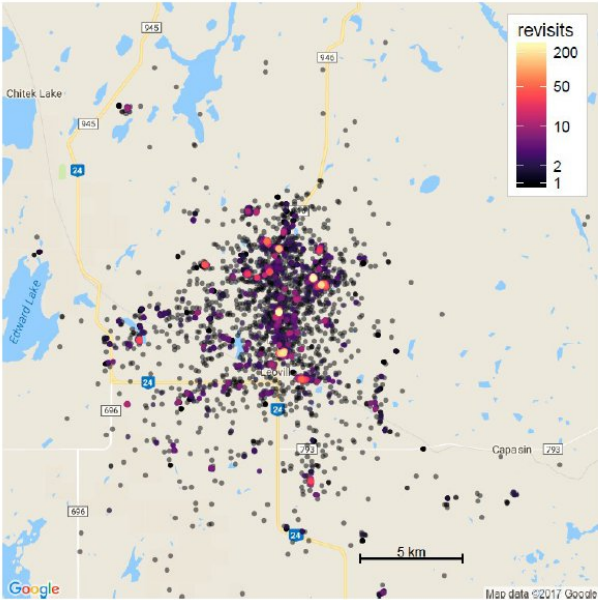
```
> habibaVisits <- getRecursions(Habiba,  
                                radius = 500,  
                                timeunits = "hours")  
> head(habibaVisits, 12)
```

	id	x	y	coordIdx	visitIdx	entranceTime
1	1	-3200.033	-386.8503	1	1	2014-01-23 11:30:42
2	1	-3200.033	-386.8503	1	2	2014-01-24 09:37:51
3	1	-3200.033	-386.8503	1	3	2014-01-24 10:46:01
4	1	-3200.033	-386.8503	1	4	2014-01-25 04:28:43
5	1	-3200.033	-386.8503	1	5	2014-01-25 10:53:12
6	1	-3200.033	-386.8503	1	6	2014-01-26 04:44:35
7	1	-3200.033	-386.8503	1	7	2014-01-26 14:07:15
8	1	-3200.033	-386.8503	1	8	2014-02-06 05:28:54
9	1	-3187.802	-359.4408	2	1	2014-01-23 11:30:42
10	1	-3187.802	-359.4408	2	2	2014-01-24 09:29:19
11	1	-3187.802	-359.4408	2	3	2014-01-25 04:29:28
12	1	-3187.802	-359.4408	2	4	2014-01-25 10:47:58

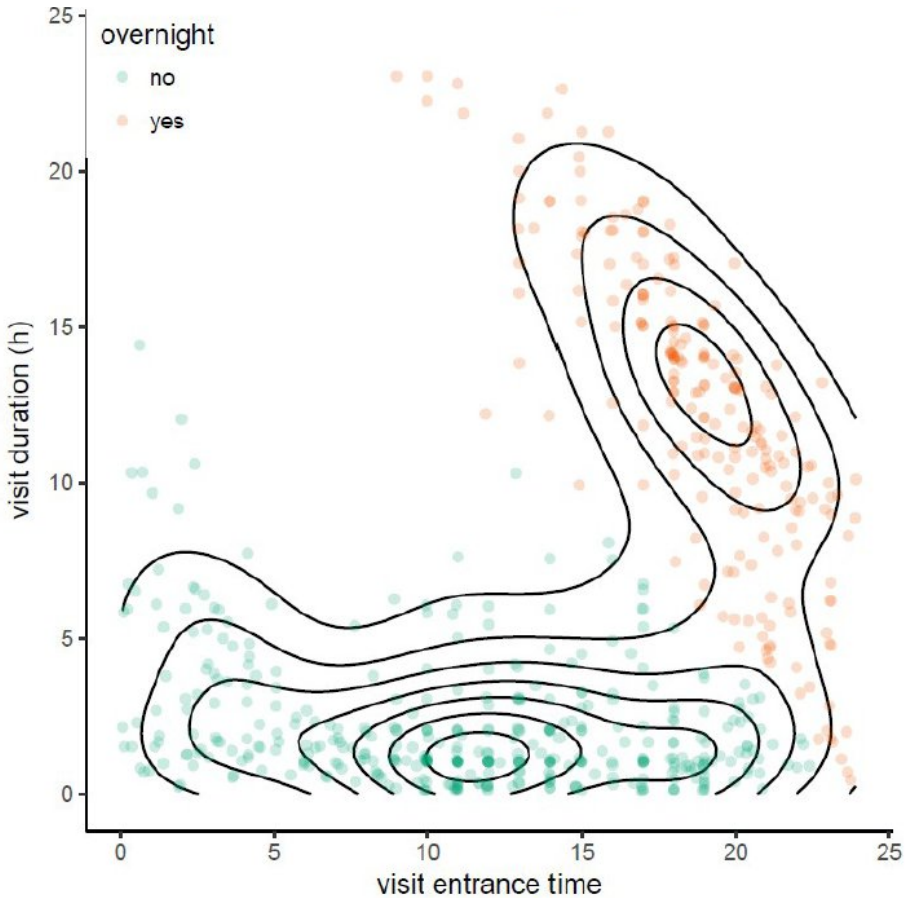
Recursions



Look at sites with > 75 visits, i.e. nests or roosts: **5 sites**



Longer visits only happen at night and include overnight (roosting sites)



Hands on

How to do this in R in the script:
“6B_TrajectoryAnalysis_Recurse.R”